

Inseok Kong

Undergraduate Researcher · Department of GeoInformatics & AI · University of Seoul

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RESEARCH INTERESTS

I am broadly interested in **Trustworthy Artificial Intelligence**, with a primary focus on **Computer Vision**. My research has primarily centered on **Test-Time Adaptation (TTA)** to enable vision models to reliably adapt to real-world distribution shifts and environmental corruptions. Recently, I am expanding my research focus toward **Uncertainty Calibration** and object hallucination mitigation in vision-language models, striving to build reliable, well-calibrated, and robust visual intelligence systems.

EDUCATION

University of Seoul

B.S. in GeoInformatics and Artificial Intelligence (Double Major)

Mar 2022 – Present

Seoul, South Korea

• **Cumulative GPA:** 3.8 / 4.5

• **Relevant Coursework:** Machine Learning, Deep Learning, Pattern Recognition, Data Structures, Algorithms.

PUBLICATIONS & PREPRINTS

[C: Conference, S: Submitted]

- [C.1] **APC: Transferable and Efficient Adversarial Point Counterattack for Robust 3D Point Cloud Recognition** Jun 2026
Geunyoung Jung, Soohong Kim, Inseok Kong, and Jiyoung Jung Denver, CO, USA
IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings (CVPR Findings)
- [C.2] **MAMVI: 3D Test-Time Adaptation via Masked Multi-View Point Clouds** Aug 2026
Inseok Kong, Geunyoung Jung, and Jiyoung Jung Lyon, France
IEEE/IAPR International Conference on Pattern Recognition (ICPR) (Oral Presentation)
- [S.1] **SATOR: Backpropagation-Free 3D Test-Time Adaptation via Sample-Adaptive Token Routing** Submitted
Inseok Kong, Geunyoung Jung, and Jiyoung Jung
Under Review
- [S.2] **Mitigating Feature Collapse for Test-Time Adaptation of 3D VLMs via Distribution-Guided Regularization** Submitted
Geunyoung Jung, Inseok Kong, and Jiyoung Jung
Under Review

RESEARCH EXPERIENCE

Statistical Inference and Information Theory Lab (SIIT)

Undergraduate Research Student (Advisor: Prof. Junmo Kim)

Jul 2026 – Aug 2026

KAIST, Daejeon, South Korea

- Conducting individual research on **Large Vision-Language Models (LVLMs)**, focusing on mitigating object hallucination and improving uncertainty calibration to address visual-textual misalignment and enhance multimodal trustworthiness.

Robotics and Computer Vision Lab (RCV)

Undergraduate Research Student (Advisor: Prof. Jiyoung Jung)

Jul 2025 – Jun 2026

University of Seoul, South Korea

- Researched **3D Test-Time Adaptation (TTA)** under severe domain shifts, investigating lightweight online adaptation strategies for 3D point cloud networks to maintain high inference accuracy while eliminating backpropagation overheads.
- Investigated **Point Cloud Robustness & 3D Adversarial Defense**, studying structural vulnerabilities of 3D point cloud classification architectures and exploring input-level defense mechanisms to neutralize point perturbations dynamically.
- Explored robust representation alignment for **3D Vision-Language Models (VLMs)**, addressing feature collapse and cross-modal misalignment issues under environmental corruptions and out-of-distribution (OOD) scenarios.

PROJECTS

LG Aimers 8th Cohort

Track: LLM Compression

Jan 2026 – Feb 2026

Online

- Researched and applied model compression & optimization techniques (pruning, quantization) on EXAONE Large Language Models.
- Optimal Autonomous Shuttle Route Planning**
Capstone Design Project & Seoul Big Data Campus

Oct 2025 – Jan 2026
Seoul, South Korea
- Developed optimal route planning algorithms using mobility big data to support customized autonomous shuttle services for transportation-vulnerable elderly citizens in Seoul.
- Sportier (UOS Entrepreneurship Club)**
Lead Developer & President (Tools: React.js, JavaScript)

Mar 2025 – Jul 2025
Seoul, South Korea
- Led the entrepreneurship club as president and developed a sports matchmaking web application featuring location-based matching and competitive tier rankings.
- Football Big Data Analytics**
Football Big Data Camp

Feb 2025
Seoul, South Korea
- Investigated practical applications of LLMs and Retrieval-Augmented Generation (RAG) in football performance analytics.

HONORS & AWARDS

- Excellence Award**
Seoul RISE AI Competition (Project: SATOR)

Jun 2026
Seoul, South Korea
- Grand Prize (Seoul Mayor Award)**
Seoul Metropolitan Big Data Campus Competition

Oct 2025
Seoul, South Korea
- Excellence Award**
UOS Capstone Design Project

Jan 2026
Seoul, South Korea
- Encouragement Award**
AICOSS Industry-Academia Hackathon (Diffusion & LLM Storybook)

Aug 2025
Seoul, South Korea
- Dean’s Award of College of Engineering**
AICOSS Deep Learning Bootcamp (2D ViT Classification)

Jan 2025
Seoul, South Korea

VOLUNTEER & MENTORING

- KB Raschool 3rd Generation Mentor**
Samsung Sori-Saem Welfare Center

Mar 2025 – Feb 2026
Seoul, South Korea
- Provided tailored mathematics mentoring and academic counseling for hearing-impaired 9th-grade students.
- Seoul Donghaeng Project Mentor**
Manna Regional Children’s Center

Sep 2024 – Dec 2024
Seoul, South Korea
- Conducted mathematics mentoring and high school entrance guidance for multicultural 8th-grade students.

SKILLS, CERTIFICATIONS & LANGUAGES

- **Technical Skills:** Python, PyTorch, MATLAB, L^AT_EX, Git, Linux / Ubuntu
- **SQL Developer (SQLD)**, Korea Data Agency (K-DATA) Dec 2024
- **Advanced Data Analytics Semi-Professional (ADsP)**, Korea Data Agency (K-DATA) Nov 2024
- **English:** TOEIC (835) Aug 2022 (Expired)

ADDITIONAL INFORMATION

- **Military Service:** Republic of Korea Army (Sergeant, Honorably Discharged) Feb 2023 – Aug 2024
- **Interests:** Formula 1 (Red Bull Racing / Max Verstappen)